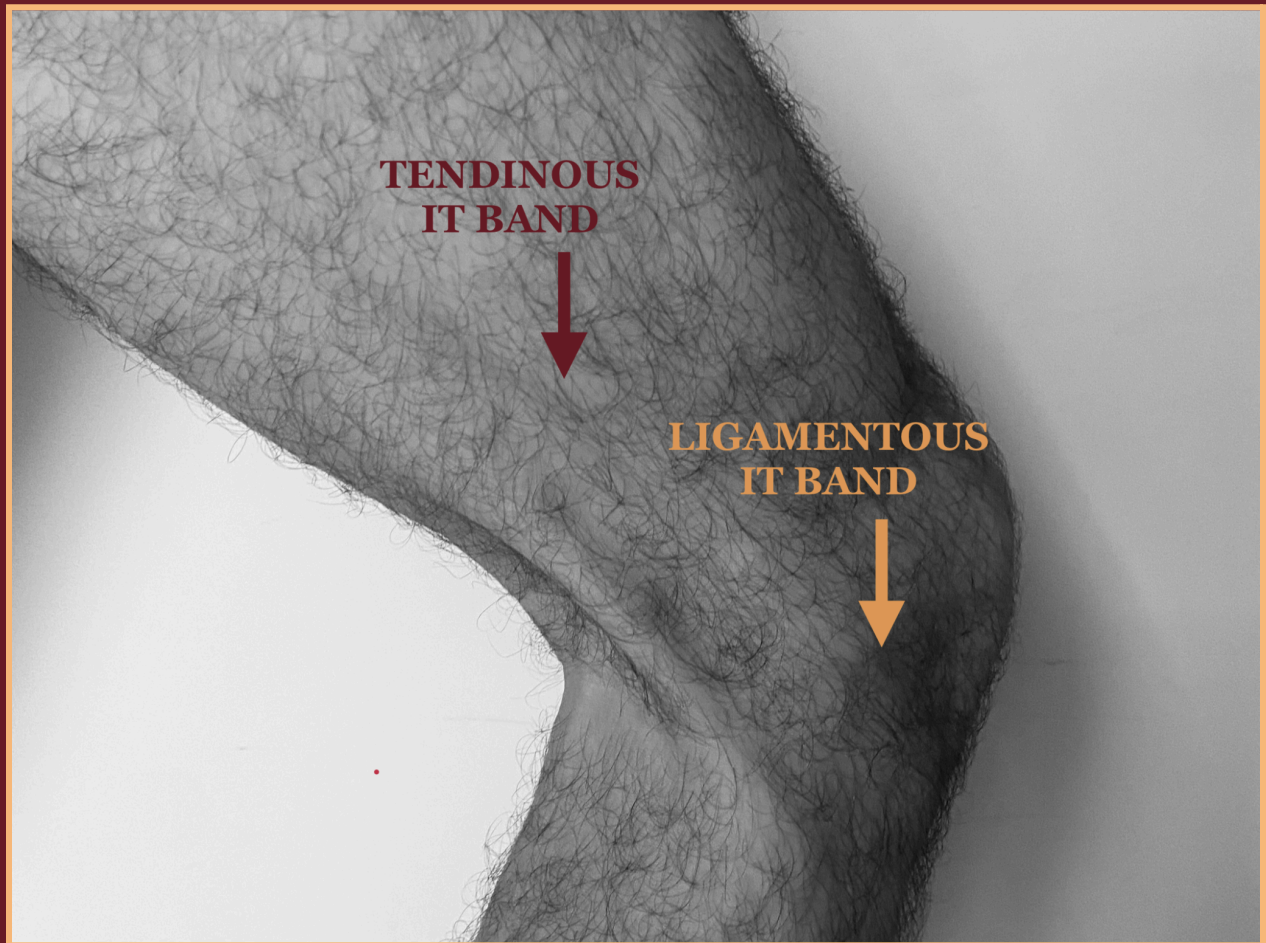


IT BAND SYNDROME

Can Massage Therapy Help Lateral Knee Pain in Runners?



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Massage therapy is excellent as a treatment for reducing pain, improving comfort during movement, and addressing tightness in the muscles that influence **iliotibial (IT) band** loading, particularly the **tensor fasciae latae (TFL)** and **gluteus medius** muscles, and it works best as part of a holistic recovery plan including temporary training modification, hip-strengthening exercises, and anti-inflammatory measures.

WHAT IS ILIOTIBIAL BAND SYNDROME?

Endurance runners, cyclists, athletes, hikers, and anyone who spends a lot of time on their feet may commonly experience a sharp pain on the outside of one or both knees while running or walking. Typically with **IT band syndrome**, no pain is felt at rest and the pain is more intense when going down-hill or when descending staircases. This pain is usually mild to start but can progress if it's ignored and can eventually become a serious obstacle to work or training.¹

Misconceptions

Before I started poring over medical meta-analyses about the IT band, I thought I had a pretty good idea of what it was and how it worked. In fact, I think most people with a hobbyist's amount of knowledge would agree that the IT band is a tight rope of fibrous connective tissue (like a tendon) spanning the length of the lateral thigh from the hip down across the knee to the head of the shin bone. These same people would also likely attribute IT band syndrome to friction from the IT band rubbing against the bony protrusion (the **lateral femoral epicondyle**) on the side of the knee. *This turns out to be pretty inaccurate.*

Anatomy

The entire musculoskeletal system is wrapped in layers of fibrous connective tissue called **fascia**. Deep fascia surrounds muscles and blends with tendons, ligaments, periosteum, and other connective tissues. Although these tissues differ in composition and function, they are mechanically continuous throughout the body, forming something like a nylon body suit, covering and penetrating every part of the musculoskeletal system.²

Similarly, *the IT band is not discrete*. It is a thicker and more raised part of the **fascia lata**, a wide sleeve of fascia covering the entirety of the lateral leg.¹

Pathology

While the IT band certainly crosses the knee joint, friction from rubbing against the lateral femoral epicondyle is anatomically unlikely because *the IT Band is continuous with the periosteum of the femur* and the **patellar retinacula** (the ligaments which keep the tendons crossing the knee from lifting away from the bone.) Because the IT band anchors to the femur in the vicinity of the lateral femoral epicondyle, it is conceptually justified to consider the proximal part of the IT band to be tendinous and the part that is distal to the epicondyle to be ligamentous.

Therefore, current evidence suggests the irritation is more likely due to compression than friction. In actuality, this irritation comes from the compression of dense, fatty, highly innervated tissue that is sandwiched between the IT band and the femur in the region of their fibrous attachments when the IT band draws towards the lateral femoral epicondyle probably as a consequence of passive tibial rotation during flexion and extension of the knee. *Repeated compression of this tissue causes inflammation and pain.*¹

HOW IS ITBS TREATED?

This condition is highly treatable through conservative nonsurgical measures. Ice and rest are likely the most essential, but massage, stretching, and targeted exercises are all effective treatment options. Anti-inflammatory medication is also a proven treatment for IT band syndrome.³

Massage Therapy

Under normal circumstances, myofascial release techniques over the IT band itself can address adhesions between the fascia lata and the **vastus lateralis**. Often, massage therapists will erroneously employ this same technique with the goal of “lengthening” a tight IT band. *The fascia of the IT band cannot be lengthened* and deep aggressive pressure directly over the irritated section of the IT band can exacerbate inflammation.

Embedded within the fascia lata is a muscle called the **tensor fasciae latae (TFL)**. Regular distance running can cause the TFL to become hypertonic and proper massage may reduce muscle tone, improve pain, and temporarily improve movement quality ultimately providing relief to the IT band.³

Similarly, trigger points in the gluteus medius and gluteus maximus muscles of the posterior hip can be released with deep accurate pressure improving hip movement and reducing excessive loading of the lateral knee so that the TFL is not relied on as heavily.⁴

Massage of the leg in general may reduce pain, improve comfort during movement, and help maintain mobility during rehabilitation, and although clinical evidence for friction massage and other modalities in this specific region remains limited, in my own practice clients have reported temporary symptom relief from a variety of techniques.

In short, properly massaging:

- the TFL
- the gluteus medius
- the gluteus maximus
- the lateral thigh and knee

can help treat ITBS by:

- reducing pain
- improving comfort during movement
- addressing hypertonicity
- supporting a rehabilitation program

Behavioral Corrections

Many runners with IT band pain also have weakness or poor control of the hip muscles that help stabilize the pelvis during running. At home, certain augmentations to training routines should be included between treatments for long-term recovery and prevention of IT Band Syndrome. Icing the inflamed lateral knee and temporarily limiting run distance are going to be the best short-term recovery strategy at home.

For long-term relief and prevention, strengthen the hip abductors and gluteal muscles while improving neuromuscular control by substituting exercises like:

- side-lying leg raises
- clamshells
- monster walks with a band
- single-leg bridges
- step-downs

in place of endurance running training. New running shoes can correct any over-pronation that might be contributing to the development of ITBS. Long-term *recovery depends on correcting the movement patterns and training factors that contributed to the problem*, rather than simply treating the painful area.⁴

WHAT CAN ATLAS BODYWORK DO?

My work focuses on the muscles and movement patterns that commonly contribute to IT band irritation rather than aggressively digging into the painful area itself. Sessions include deep tissue massage of the hip and leg, along with guidance on activity modification and exercises.

References

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